USING ACTIVATION FUNCTIONS FOR IMPROVING MEASURE-LEVEL AUDIO SYNCHRONIZATION

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ABSTRACT

Audio synchronization aims at aligning multiple recordings of the same piece of music. Traditional synchronization approaches are often based on dynamic time warping using chroma features as an input representation. Previous work has shown how one can integrate onset cues into this pipeline for improving the alignment's temporal accuracy. Furthermore, recent work based on deep neural networks has led to significant improvements for learning onset, beat, and downbeat activation functions. However, for music with soft onsets and abrupt tempo changes, these functions may be unreliable, leading to unstable results. As the main contribution of this paper, we introduce a combined approach that integrates activation functions into the synchronization pipeline. We show that this approach improves the temporal accuracy thanks to the activation cues while inheriting the robustness of the traditional synchronization approach. Conducting experiments based on string quartet recordings, we evaluate our combined approach where we transfer measure annotations from a reference recording to a target recording.

1. INTRODUCTION

In music information retrieval (MIR), synchronization techniques are essential for several applications including score following [1], content-based retrieval [2], automatic accompaniment [3], or performance analysis [4, 5]. Beside these applications, music synchronization has a great potential to simplify data augmentation, data annotation, and model evaluation. For example, one can use music synchronization to obtain additional training data for deep learning methods semi-automatically by transferring annotations from one recording to another. Furthermore, using music synchronization, one can transfer measure positions between audio recordings for navigation purposes, structural segmentation, and cross-version analysis [6, 7].

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While traditional synchronization approaches typically rely on alignment algorithms such as dynamic time warping (DTW) and conventional chroma features used as the input representation [8, 9], the integration of additional onset-related information has proven to enhance the synchronization accuracy [10–12]. Inspired by the combined approach from [12], where *decaying locally adaptive chroma onset* (DLNCO) features are integrated into the synchronization pipeline, we incorporate in this paper onset, beat, and downbeat activation functions to obtain a better temporal accuracy while retaining the robustness of the original chroma-based synchronization approach (see Figure 1 for an illustration of the overall approach). The addition of activation functions results in a grid-like structure in the cost matrix, which guides the alignment through activation cues that point to note onsets or other musical events.

While the integration of DLNCO and spectral flux (SF) have led to substantial improvement of synchronization results [12,13], the detection of soft onsets constitutes a challenging problem due to their long attack phase with a slow rise in energy. To adapt the onset detection task to music recordings which comprise soft onsets and temporal–spectral modulations such as vibrato (e.g., string music), Böck and Widmer [14] introduced the superflux (SF*) feature. Furthermore, deep learning (DL) methods such as bidirectional long short-term memory (BLSTM) networks [15] and convolutional neural networks (CNN) [16] have led to significant improvements compared to conventional onset detectors.

As the main contribution of this paper, we show how one can integrate conventional and DL-based activation functions into the synchronization pipeline. Different from the approach in [12], we do not apply any hard peak picking but directly use onset-related activation cues. Furthermore, we go beyond onsets by integrating activation functions that indicate onset, beat, and downbeat positions. In particular for music with noisy and unreliable onset cues, we show that beat and downbeat cues are more reliable and better suited for improving the synchronization accuracy. For extracting beat and downbeat activation functions, we build on recent work by Böck et al. [17, 18], using recurrent neural network (RNN) models for extracting beat and downbeat activation functions.

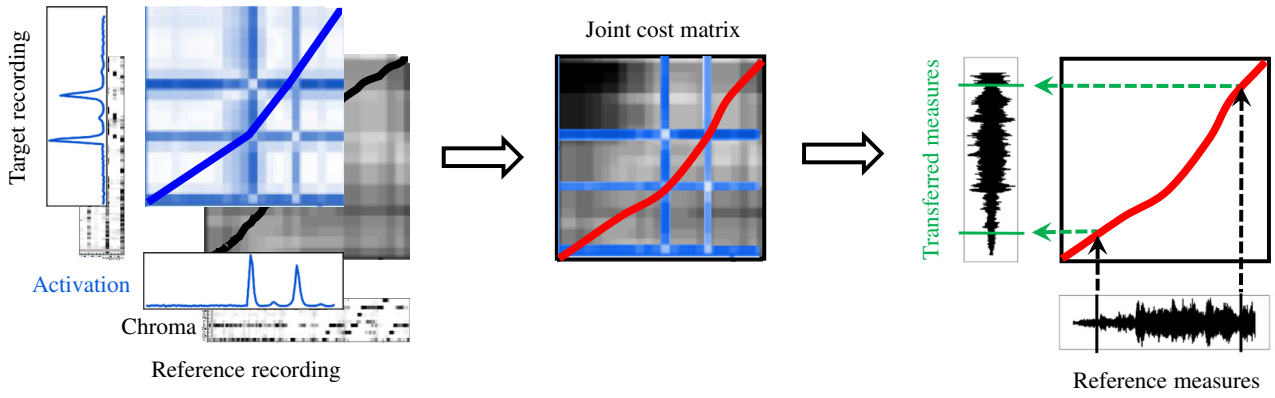


Figure 1: Overview of the combined approach integrating activation functions into a conventional chroma-based synchronization pipeline. The cost matrix computed with activation cues (blue) yields a grid-like structure to guide the alignment of musical events, whereas the chroma-based cost matrix (black) accounts for the robustness of the overall synchronization. The resulting warping path (red) is used for transferring measure positions.

To better understand our improved synchronization pipeline, we compare several synchronization approaches where we transfer measure annotations from a reference recording to a target recording, similar to [19]. In particular, we conduct systematic experiments based on three versions of the String Quartet No. 12 in F major, Op. 96, composed by Antonín Dvořák. As string music generally comprises vibrato, tremolo, rubato, and abrupt tempo changes, which increase the musical complexity, the synchronization of string quartets is a challenging scenario. We show that integrating DL-based activation functions significantly improves the temporal accuracy while retaining the robustness of chroma features.

The remainder of the paper is organized as follows. In Section 2, we introduce our combined synchronization approach, explore conventional and DL-based activation cues, and show how to integrate activation functions into the synchronization pipeline. In Section 3, we present our dataset, the measure transfer between string quartet recordings as our application scenario, and report on our systematic experiments and empirical results. Finally, we conclude in Section 4 with prospects on future work.

2. COMBINED SYNCHRONIZATION APPROACH

In this section, we show how activation functions can be integrated into the synchronization pipeline to enhance the temporal accuracy of traditional chroma-based synchronization approaches. Here, we regard the activation function as a function that yields a value between 0 and 1. Each entry in the function indicates the likelihood of a certain musical event, e.g., onsets, beats, or downbeats, for each frame (time position). In the ideal case, the value of the activation function is one when an event occurs and zero otherwise. Note that we do not apply any peak picking in our approach, but only use the activation functions as temporal cues. This is opposed to onset detection or beat tracking where one needs to apply a temporal decoding method to obtain an explicit representation of onset and beat positions from the activation functions.

In the following, we first explore conventional onset-based activation functions in Section 2.1. Then, in Section 2.2, we investigate DL-based onset, beat and downbeat detectors. Finally, in Section 2.3, we explain how we integrate activation functions into the synchronization pipeline.

2.1 Conventional Onset-Based Activation Functions

2.1.1 DLNCO

DLNCO features are 12-dimensional pitch-based onset features, which combine the robustness of the chroma features with the accuracy of one-dimensional onset features. To compute DLNCO features, we first apply a pitch-wise audio decomposition. Then, we derive pitch-wise onset cues by considering points of energy increase (see Figure 2c for an illustration). DLNCO features are particularly suited for the music with clear note attacks such as piano music. For further details about the computation of DLNCO features, we refer to [12].

2.1.2 SF

Spectral flux (SF) captures the changes in the spectral content of an audio signal, and is widely used for onset detection [9, 20]. For the computation of SF, we apply a first-order differentiator on the log-compressed magnitude spectrogram of a music recording. Half-wave rectification follows the differentiation to keep only the positive differences between subsequent frames. As a final step, we subtract a local average function to enhance the peak structure (see Figure 2d).

2.1.3 SF*

Superflux (SF*) is a modified version of SF for detecting soft onsets [14]. These features are suitable for music recordings with vibrato, such as strings quartets. Similar to

the SF algorithm, SF* also relies on the detection of positive changes in the energy over time. However, it includes a trajectory-tracking stage through maximum filtering, instead of simply calculating the difference between spectral bins over time. Trajectory tracking helps to suppress spurious spectral peaks, especially arising from vibrato. For further information, we refer to [14] (see also Figure 2e).

2.2 DL-Based Activation Functions

2.2.1 CNN Onset Detector

Schlüter and Böck [16] approach the onset detection task as a computer vision problem, where magnitude spectrograms of the audio recordings are used as the input to a CNN. Onsets are often characterized by rapid transient changes in the spectrum, resulting in sharp edges that are clearly visible in a spectrogram. Using convolutional kernels, one can easily detect these sharp edges of onsets. Similar to SF-based methods, the proposed CNN model computes spectro–temporal differences and captures percussive and pitched onsets. The resulting activation function is referred to as `DL-O` (see Figure 2f for an example).

2.2.2 RNN Beat Detector

In the case of unreliable and noisy onset cues, using beat activation functions constitutes a more feasible solution to improve the temporal alignment. To compute such activation functions, we use the BLSTM model by Böck and Schedl [17] for framewise beat detection. BLSTMs can effectively model the temporal context of the data and is therefore suitable for beat tracking. In the proposed approach, magnitude spectrograms computed with three different window lengths, and their first order differences are used as the input to the network. The network outputs encode the likelihoods of beat positions, as illustrated by Figure 2g. In our experiments, the resulting beat activation functions are denoted as `DL-B`.

2.2.3 RNN Downbeat Detector

Böck et. al [18] present an RNN model to jointly detect beat and downbeats. Like the previously mentioned onset and beat detectors, this model also operates on magnitude spectrograms. The downbeat detector uses an RNN similar to the proposed network in [17] to model beats and downbeats. In our experiments, we only use the probability of downbeats as the activation cues, which we denote as `DL-D` (see Figure 2h for an illustration).

2.3 Combined Synchronization with Activation Functions

To find the optimal alignment between two feature sequences $X := (x_1, \ldots, x_N)$ and $Y := (y_1, \ldots, y_M)$, where $n \in \{1, \ldots, N\}$ and $m \in \{1, \ldots, M\}$, we rely on DTW. By comparing each pair of elements in the feature sequences, we obtain a cost matrix $\mathbf{C}(n, m) := c(x_n, y_m)$ of size $N \times M$, where c defines a local cost measure. Then, an *optimal warping path* is determined via dynamic programming. We refer to [9] for a detailed account on DTW

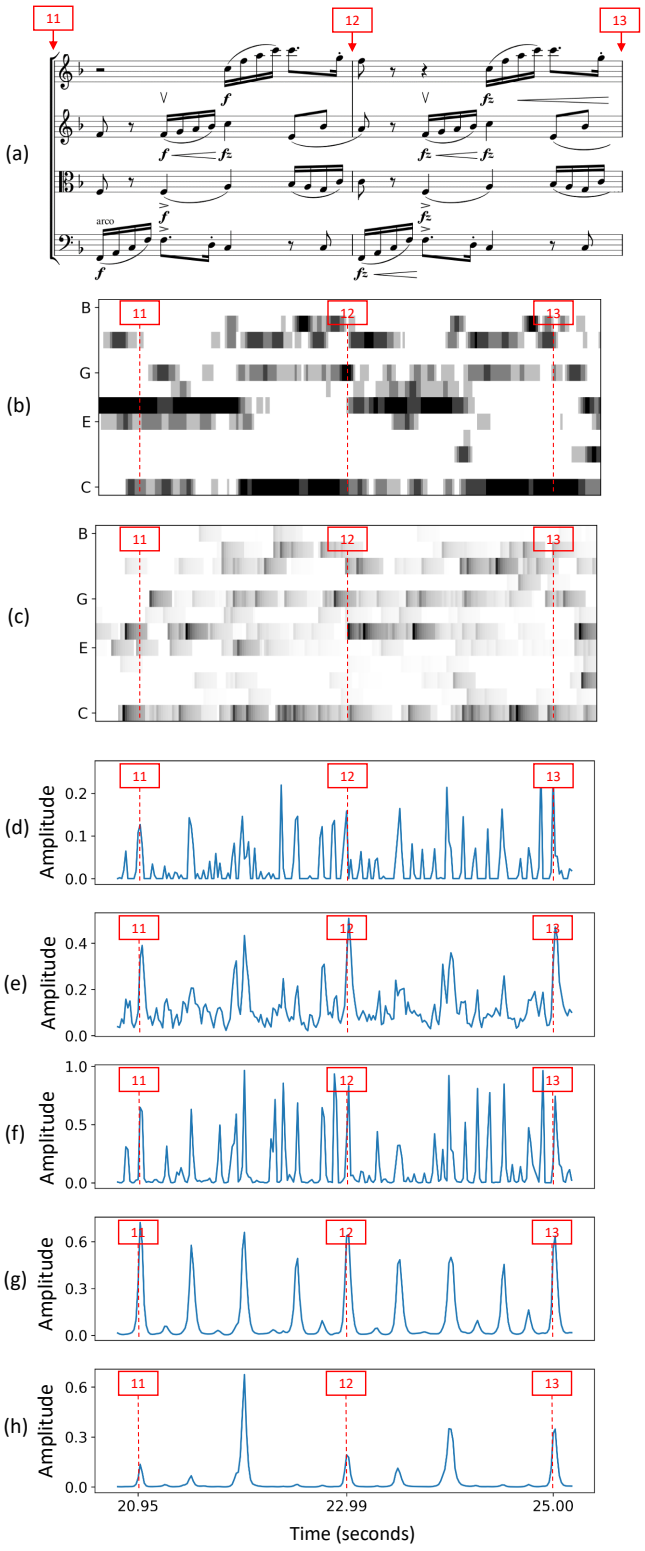


Figure 2: Chroma features and activation functions computed for an excerpt of the String Quartet No. 12 in F major, Op. 96 (first movement) composed by Antonín Dvořák, performed by the Borromeo Ensemble. Activation functions are shown in blue and ground-truth measure positions in red. **(a)** Sheet music representation of the measures 11–13. **(b)** Chroma **(c)** DLNCO **(d)** SF **(e)** SF* **(f)** Onsets (`DL-O`) **(g)** Beats (`DL-B`) **(h)** Downbeats (`DL-D`)

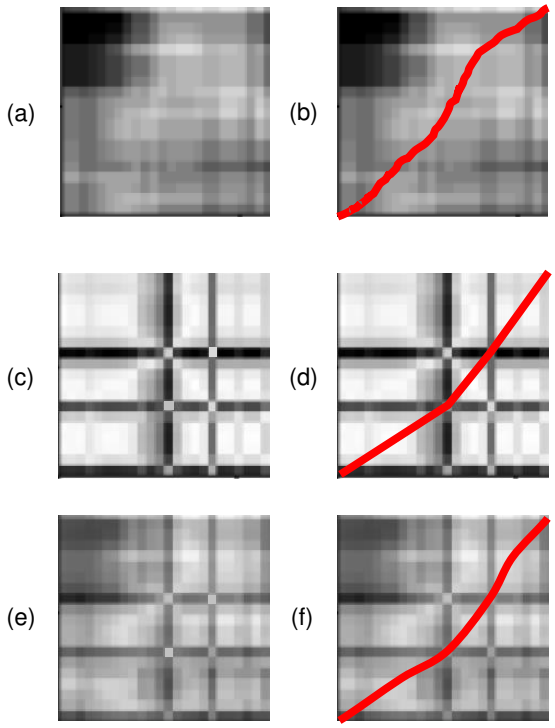


Figure 3: Excerpts from cost matrices and corresponding warping paths computed with DTW **(a) / (b)**: $\mathbf{C}_{\texttt{CHROMA}}$, **(c) / (d)**: $\mathbf{C}_{\texttt{ACT}}$, **(e) / (f)**: $\alpha\mathbf{C}_{\texttt{CHROMA}} + (1-\alpha)\mathbf{C}_{\texttt{ACT}}$, with $\alpha = 0.5$.

for music synchronization. For efficient implementations, we refer to [21, 22].

We now adapt the combined synchronization idea by Ewert et al. [12], integrating conventional onset-based and DL-based activation functions into the synchronization pipeline. Building upon this approach, we introduce three cost matrices. The first one $\mathbf{C}_{\texttt{CHROMA}}$ is a cost matrix based on the normalized chroma features and the cosine distance, see Figure 3a for an example. The second cost matrix $\mathbf{C}_{\texttt{ACT}}$ is computed using the Euclidean distance and conventional onset-based or DL-based activation functions as introduced in Section 2.1 and Section 2.2, respectively. Note that $\mathbf{C}_{\texttt{ACT}}$ exhibits grid-like structures. The visible horizontal and vertical grid lines of high cost (shown in black) correspond to high values in the first and second function, respectively. Only when a horizontal grid line intersects with a vertical grid line, the cost matrix has a small cost value at this intersection point (and also in a small neighborhood). This is where a high activation value of the first sequence meets another high activation value of the second sequence. In other words, these *intersection cells* encode a pair of time positions where two musical events (onsets, beats, downbeats) meet (see Figure 3c). Furthermore, the sections in the activation functions which have low values lead to homogeneous, zero-cost regions in the cost matrix $\mathbf{C}_{\texttt{ACT}}$. The third matrix is the sum of two cost matrices

$$\mathbf{C} = \alpha\mathbf{C}_{\texttt{CHROMA}} + (1-\alpha)\mathbf{C}_{\texttt{ACT}}, \qquad (1)$$

where $\alpha \in [0, 1]$ is a weighting parameter. The sum $\mathbf{C}$ accounts for both harmonic or melodic information of the representations via $\mathbf{C}_{\texttt{CHROMA}}$ and additional activation cues via $\mathbf{C}_{\texttt{ACT}}$. Figure 3e illustrates an example using $\alpha = 0.5$.

Comparing the resulting optimal warping paths using $\mathbf{C}_{\texttt{CHROMA}}$ in Figure 3b, $\mathbf{C}_{\texttt{ACT}}$ in Figure 3d, and $\mathbf{C}$ in Figure 3f, we can observe an enhancement of the temporal alignment. The inclusion of DL-based onset, beat, and downbeat cues leads to an improvement of the warping path guided by grid structure's intersection points. Note that $\mathbf{C}_{\texttt{ACT}}$ remains zero in the regions without any novel events, and the overall alignment of $\mathbf{C}$ is mainly guided by $\mathbf{C}_{\texttt{CHROMA}}$.

3. EXPERIMENTS

3.1 Dataset

The genre of string quartet is composed for a small conductor-less ensemble, which consists of two violins, a viola and a violoncello. In our experiments, we use three versions (performances) of the String Quartet No. 12 in F major, Op. 96, by Antonín Dvořák, which comprises four movements. To give an insight of the musical properties of the string quartet, Table 1 provides the number of measures, time signature, and global tempo for each movement based on the recordings in our dataset. Note that the global tempo does not reflect any local tempo deviations. Its purpose is to indicate at what pace (average of three performances) a given movement is performed. In each recording, the repetitions are played as notated in the sheet music, thus ensuring structural consistency.

Movement	#Measures	Time signature	Global Tempo
M1	239	4/4	100
M2	97	6/8	84
M3	244	3/4	185
M4	382	2/4	140

Table 1: Overview of four movements, including number of measures, time signature, and global tempo in BPM for each movement.

For each recording (in the following referred to as version), we manually annotated the measure positions. In Table 2, the name of the version, the identifier, the recording year for each version, and the duration of each movement are listed. Note that each of the three performances last around 26 minutes in total, whereas durations of the movements may vary across different versions.

			Duration (seconds)				
Version	ID	Year	M1	M2	M3	M4	Σ
Alban Berg	A	1991	599	410	238	323	1570
Borromeo	B	2012	540	465	228	338	1571
Prague	P	1973	584	424	250	322	1580
		Σ	1723	1299	716	983	4721

Table 2: Version, identifier, recording year, and duration of each movement.

	DL-B & DBN			DL-D & DBN		
	P	R	F	P	R	F
M1	0.194	0.806	0.312	0.648	0.586	0.612
M2	0.138	0.820	0.236	0.657	0.643	0.650
M3	0.327	0.539	0.407	0.524	0.224	0.314
M4	0.517	0.908	0.655	0.876	0.647	0.742
ϕ	0.294	0.768	0.403	0.676	0.525	0.580

Table 3: Precision (P), Recall (R), and F-measure (F) based on methods DL-B for beat, and DL-D for downbeat tracking with DBN post-processing, evaluated on reference measure annotations and tolerance $\tau = 70$ ms. ϕ denotes the average accuracy over four movements M1, M2, M3, and M4.

	$\tau = 30$ ms		$\tau = 70$ ms		$\tau = 100$ ms	
	CHROMA	DL-OBD	CHROMA	DL-OBD	CHROMA	DL-OBD
M1	0.408	0.576	0.723	0.813	0.817	0.877
M2	0.396	0.600	0.694	0.842	0.789	0.917
M3	0.528	0.655	0.827	0.912	0.910	0.956
M4	0.485	0.662	0.773	0.884	0.861	0.930
ϕ	0.454	0.624	0.754	0.863	0.844	0.920

Table 4: Accuracy values based on chroma and DL-OBD for different tolerances $\tau = 30,\ 70,\ 100$ ms. ϕ denotes the average accuracy over four movements M1, M2, M3, and M4.

3.2 Beat and Downbeat Tracking

As a baseline, we first introduce how DL-based beat [17] and downbeat trackers [18] perform in the detection of measure positions, where a dynamic Bayesian network (DBN) was used for post-processing (peak picking). Table 3 provides precision, recall, and F-measure values using a tolerance of $\tau = 70$ ms. Here, each entry indicates the mean value over different performances. Due to the higher density of beats, the beat tracker reveals a higher recall than the downbeat tracker for each movement, leading to a low precision and F-measure. Furthermore, each movement has a different time signature, which results in a different number of beats per measure and needs to be taken into consideration as prior information for the DBN.

3.3 Synchronization Results

In this section, we describe our experimental setting and evaluate audio alignments obtained using chroma features and different activation functions. In our experiments, we use the resulting warping path to transfer the measure positions annotated for the reference recording to the target recording. Given two versions of the same music piece with the time-continuous axes $[0, T_1]$ and $[0, T_2]$, the monotonous alignment can be modeled as a function

$$\mathcal{A} : [0, T_1] \to [0, T_2].$$

The pairwise alignment error ϵ_P for a given alignment of two recording is specified as the mean over the values

$$\epsilon_P(g_1) := |\mathcal{A}(g_1) - g_2|,$$

where $(g_1, g_2) \in [0, T_1] \times [0, T_2]$ denotes the ground-truth pairs of measure annotations. As an evaluation metric, we use *accuracy*, which is defined as the proportion of correctly transferred measure positions with a pairwise alignment error below a given tolerance τ [23].

In the following, we use eight different synchronization approaches based on conventional chroma features and the combination of chroma features with DLNCO features, SF, SF*, DL-based onsets (DL-O), DL-based beats (DL-B), DL-based downbeats (DL-D), and finally a 3-dimensional stacked activation function combining DL-based onsets, beats, and downbeats (DL-OBD) (see Section 2 for a detailed overview of activation functions). We use a feature rate of 50 Hz for the computation of chroma, and conventional onset features. To generate DL-based features, we use the *madmom* [24] library, for which we utilize the default setting 100 Hz as the feature rate, and downsample generated features to 50 Hz (after low-pass filtering).

3.3.1 Overall Result

To get a first impression of the alignment behavior of different approaches, Figure 4 illustrates an overview of average accuracy values (averaged over all movements and all pairs of different performances) and for different tolerances τ. Obviously, one can observe that the synchronization accuracy improves with increasing threshold. For example, using $\tau = 30$ ms, the average synchronization accuracy is 0.454 when using only chroma features, and the accuracy increases to 0.624 when integrating the DL-OBD activation cues. This is a substantial improvement.

Next, we focus on the comparison of different approaches for $\tau = 70$ ms, which is a common tolerance value for the evaluation of music synchronization and beat tracking procedures. The inclusion of DLNCO slightly worsens the alignment since DLNCO features are not suited for soft onsets as occuring in string music. However, the integration of SF and SF* into the synchronization pipeline results in a better accuracy. Among conventional onset features, SF* shows a better performance than SF and DLNCO, owing to the fact that SF* features account for the detection of soft onsets and are therefore more suitable for string music. Furthermore, using DL-based methods leads to a better synchronization result compared to the conventional methods. Note that the integration of DL-OBD, which combines DL-O, DL-B, DL-D, reveals the best accuracy among all the synchronization approaches. It is also interesting to observe that DL-B, which is based on beats, is the second-best among DL-based approaches and leads to better accuracy values than downbeat-based DL-D, whereas DL-O reveals the lowest accuracy among the DL-based approaches.

In general, similar trends can be observed when using other thresholds. Using $\tau = 500$ ms, all synchronization approaches yield nearly perfect results.

3.3.2 Dependency on Movement

In our next experiment, we analyze the synchronization accuracy across different movements. Table 4 provides a comparison of alignment results based on the conventional chroma-based approach and our proposed combined

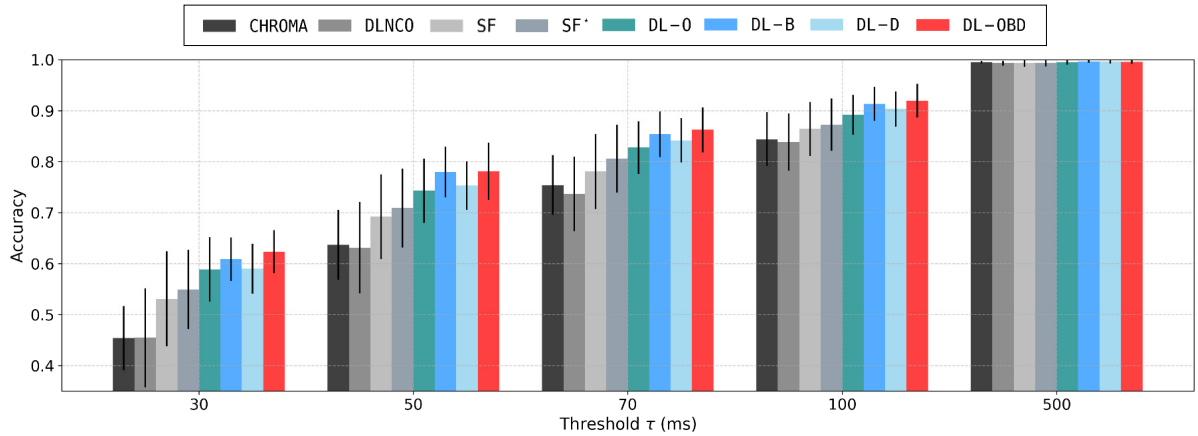


Figure 4: Comparison of the average accuracy values for different synchronization approaches and different threshold parameters τ. The accuracy denotes the proportion of correctly transferred measure positions having an error below a given tolerance τ.

	$\tau = 30$ ms		$\tau = 70$ ms		$\tau = 100$ ms	
	CHROMA	DL-OBD	CHROMA	DL-OBD	CHROMA	DL-OBD
AP	0.494	0.636	0.774	0.864	0.839	0.928
PA	0.500	0.632	0.778	0.864	0.847	0.928
AB	0.434	0.576	0.722	0.849	0.830	0.907
BA	0.451	0.580	0.743	0.863	0.857	0.920
BP	0.429	0.662	0.762	0.870	0.850	0.919
PB	0.417	0.657	0.746	0.866	0.843	0.915
ϕ	0.454	0.624	0.754	0.863	0.844	0.920

Table 5: Accuracy based on chroma and DL-OBD across different synchronization pairs, for different tolerances τ. The first column indicates the pair of versions (see Section 3.1).

method DL-OBD per movement for different tolerances τ. For example, considering the first movement and $\tau = 70$ ms, the accuracy of 0.723 for the chroma-based approach increases to 0.813 when using our combine approach DL-OBD. One can observe a similar trend across different movements for different tolerance parameters τ. The second movement tends to yield the lowest accuracy values when using only chroma features, while the integration of DL-OBD significantly improves the synchronization accuracy from 0.694 to 0.842 for the second movement. One reason may be that the beat and downbeat information leads to a significant improvement in synchronization accuracy of slower sections.

3.3.3 Dependency on Performance

As a final experiment, we provide a comparison of the accuracy values across different performances for different tolerance parameters τ in Table 5. In general, using a combination of chroma and activation functions significantly improves the accuracy for all the synchronization pairs and tolerances. For $\tau = 70$ ms, chroma reveals an average accuracy of 0.774 for AP and DL-OBD improves the accuracy to 0.864. Remarkably, across other synchronization pairs the synchronization accuracy values are similar. Nonetheless, deviations may occur due to soft onsets, slight inconsistencies in ground-truth annotations, and linear interpolation while measure transfer. Note that the synchronization accuracy rather depends on the musical complexity and structure, e.g., across different movements, but not on the performances.

4. CONCLUSION

In this article, we investigated the incorporation of conventional onset features, and activation cues obtained by recent DL-based onset, beat, and downbeat detectors to a conventional chroma-based synchronization pipeline. We showed that the integration of a combined version of onset, beat, and downbeat activation functions significantly improves the synchronization accuracy while maintaining the robustness of the original chroma-based synchronization approach. For the future, we will incorporate other features, which capture smoother note transitions for string quartets. We also aim at extending our string quartet dataset and evaluating the synchronization on beat-level annotations. Moreover, we will further investigate the role of the hyperparameter α (see Equation 1), which can be optimized as a time-dependent parameter.

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